Predictive Insight Engine

Part A — Conceptual Understanding (Theory)

1. What are Supervised Learning Algorithms?

Supervised learning is a category of machine learning in which a model is trained on a labeled dataset — one where every input (a set of features, X) is paired with a known, correct output (the target, y). The algorithm learns the mapping between X and y during training, and this learned mapping is then used to predict the output for new, unseen inputs. In this project, the model is trained on historical property records (area, bedrooms, location, etc.) paired with their known sale price, so it can later predict the price of a house it has never seen before.

2. Regression Algorithms vs. Classification Algorithms

Both are types of supervised learning, but they differ in the nature of the target variable they predict:

Aspect	Regression	Classification
Output type	Continuous numeric value	Discrete category / class label
Example	Predicting house price in INR	Predicting whether a house sells within 30 days (Yes/No)
Typical metrics	MSE, MAE, RMSE, R^2	Accuracy, Precision, Recall, F1-score
Common algorithms	Linear/Polynomial Regression	Logistic Regression, Decision Trees

This project uses **regression**, since house price is a continuous numeric quantity.

3. Simple Linear Regression

Simple Linear Regression models the relationship between a single independent variable (X) and a dependent variable (y) as a straight line:

$$y = m \cdot X + c$$

where **m** is the slope (the change in y for a one-unit change in X) and **c** is the intercept (the predicted value of y when X = 0). The algorithm finds the values of m and c that minimize the total squared difference between the actual and predicted values (the Ordinary Least Squares method). In this project, house price is predicted from house area alone using this technique in Part C.

4. Assumptions of Linear Regression

- **Linearity:** The relationship between the independent variable(s) and the dependent variable is linear.
- **Independence of errors:** Residuals (errors) are independent of each other — no autocorrelation, as in non-time-series data.

- **Homoscedasticity:** The variance of residuals is constant across all levels of the independent variable(s); the spread of errors should not grow or shrink with the predicted value.
- **Normality of residuals:** Residuals are approximately normally distributed, which supports valid confidence intervals and hypothesis tests on the coefficients.
- **No (or low) multicollinearity:** In multiple regression, independent variables should not be highly correlated with each other, or coefficient estimates become unstable and hard to interpret.

5. Bias–Variance Trade-off

Bias is the error introduced by approximating a real-world problem with a simplified model — high bias means the model is too simple and misses real patterns (underfitting). **Variance** is the error introduced by the model's sensitivity to small fluctuations in the training data — high variance means the model fits the training data (including its noise) too closely and fails to generalize to new data (overfitting). The trade-off is that reducing bias (by adding model complexity) typically increases variance, and vice versa. The goal of model selection is to find the sweet spot that minimizes total prediction error on unseen data — neither too simple nor too complex.

6. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too well — including its noise and random fluctuations — resulting in excellent training performance but poor performance on new, unseen data. Example: a very high-degree polynomial regression that curves through every single training point but oscillates wildly and predicts poorly on the test set.

Underfitting occurs when a model is too simple to capture the underlying pattern in the data, resulting in poor performance on both training and test data. Example: using Simple Linear Regression with only house area as the predictor when price actually depends on many more factors (location, age, amenities) — the model has high bias and cannot capture the real relationship even on the training set.

Prepared as Part A (Conceptual Understanding) of the Predictive Insight Engine project. See the accompanying Jupyter Notebook for the full practical implementation (Parts B–I).